

On the Statistical and Economic Fallacy of Post-Crash Mean Reversion in Indian Equities

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Abstract

We empirically investigate the core quantitative finance hypothesis: 'After a significant one-day fall in NIFTY, the market tends to recover over the next few trading days.' Utilizing 19 years of continuous, survivorship-bias-free daily data (September 2007 to September 2026, 4,664 trading bars), we demonstrate that post-shock positive returns are an optical artifact of unconditional equity market drift. For daily shocks $\leq -2.0\%$ ($N = 200$), the average 5-day post-drop return ($+0.32\%$) is statistically indistinguishable from unconditional rolling drift ($+0.21\%$), yielding an abnormal return of only $+0.10\%$ (Welch t-statistic = $+0.25$, $p = 0.8031$; 10,000-sample Circular Block Bootstrap $p = 0.7863$). Survival analysis reveals that by Day 5, the cumulative probability of fully reclaiming the drop is only 49.0%. Over 73% of large shocks occur in structural bear regimes (Price < 200 SMA) where downside momentum and volatility clustering dominate ($+0.14\%$ recovery). Net of realistic Indian statutory friction and overnight gap risk, naive dip-buying yields a negative CAGR of -0.39% with a catastrophic -46.18% maximum drawdown. The unconditioned mean-reversion hypothesis is firmly rejected.

Keywords — Event Study; Mean Reversion; NIFTY 50; Overreaction Hypothesis; Market Microstructure; Survival Analysis; Volatility Clustering; Regime Conditioning.

1. Introduction & Theoretical Foundations

A foundational doctrine in empirical finance is the Overreaction Hypothesis (De Bondt & Thaler, 1985), which posits that extreme negative price shocks induce panic selling and temporary liquidity voids, causing asset prices to overshoot fundamental equilibrium before mean-reverting. In systematic trading, this heuristic is colloquially operationalized as 'buying the dip'.

However, rigorous testing of mean-reversion in emerging market equity benchmarks such as India's NIFTY 50 introduces three fatal methodological pitfalls that are routinely overlooked in commercial research:

1.1 Unconditional Drift Conflation: Equity indices exhibit positive unconditional geometric growth. Evaluating nominal post-shock returns without subtracting identical-horizon rolling drift attributes broad equity risk premia to trading alpha.

1.2 Look-Ahead Close Execution Fallacy: Calculating forward returns from Day T Close assumes frictionless execution at 15:30:00 IST. In practice, traders must absorb overnight auction gap risk at Day T+1 Open, which extracts the majority of theoretical rebound.

1.3 Autoregressive Volatility Clustering: Following Mandelbrot (1963) and Engle (1982), large price innovations cluster chronologically, converting post-drop environments into regimes of persistent downside momentum and elevated liquidation risk.

2. Mathematical Formulation of Event Space

Let $\{S_t\}$ denote the discrete index price sequence. The 1-day percentage return is $R_t = (S_t - S_{t-1}) / S_{t-1}$. An extreme shock event is identified by the indicator:

$$I_t(\theta) = 1 \text{ if } R_t \leq \theta, \text{ } 0 \text{ otherwise,}$$

where baseline $\theta = -2.00\%$, corresponding to the 95.7th percentile left tail ($N = 200$ events). The multi-horizon forward return across horizon $h \in \{1, \dots, 20\}$ is formulated under two distinct execution models:

$$R_{t,h}^{(C)} = (S_{t+h}^{(C)} - S_t^{(C)}) / S_t^{(C)} \quad (\text{Theoretical Close}),$$
$$R_{t,h}^{(O)} = (S_{t+h}^{(C)} - S_{t+1}^{(O)}) / S_{t+1}^{(O)} \quad (\text{Realistic Open}).$$

Abnormal return is formally defined as the differential expectation over identical holding horizons: $AR_h = E[R_{t,h} | I_t = 1] - E[R_{t,h}]$, isolating genuine alpha from broad market appreciation.

3. Limits to Arbitrage & Liquidity Holes

As formalized by Shleifer & Vishny (1997) and Morris & Shin (2004), liquidity provision during severe market declines is constrained by capital mandates, margin calls, and risk-parity de-leveraging. Rather than stepping in to buy undervalued assets, institutional desks withdraw liquidity, inducing autocorrelation in downside price moves and creating persistent post-drop drawdown traps.

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4. Data Integrity & Verification Pipeline

We utilize continuous daily OHLCV series for the NIFTY 50 index spanning September 17, 2007 through September 21, 2026 (4,664 trading sessions). An automated data integrity pipeline enforces: (1) strict chronological monotonic ordering, (2) zero-tolerance rejection of non-positive prices or volumes, and (3) structural candlestick envelope invariance: $High_t \geq \max(Open_t, Close_t)$ and $Low_t \leq \min(Open_t, Close_t)$. The series is fully survivorship-bias adjusted.

5. Baseline Benchmarking & Statistical Inference

To test $H_0: AR_h = 0$ against $H_1: AR_h > 0$, we construct the Unconditional Baseline Distribution across all 4,659 rolling 5-day windows. We deploy two-sample Welch t-tests with Satterthwaite degrees of freedom:

$$t = (X_{\text{event}} - X_{\text{base}}) / \sqrt{(s_1^2/N_1 + s_2^2/N_2)}, \quad \nu \approx (s_1^2/N_1 + s_2^2/N_2)^2 / [\sum (s_i^2/N_i)^2 / (N_i - 1)].$$

We also execute a 10,000-sample Circular Block Bootstrap (Politis & Romano, 1994) with optimal block length $b = \lceil n^{1/3} \rceil = 17$. Pseudo-time series are sampled via overlapping blocks $B_j = (R_{t_j}, \dots, R_{t_j+b-1})$ with circular wrap-around to preserve stationary serial dependence.

Table 1. Forward Return Distribution vs Unconditional Baseline ($N = 200$)

Horizon (h)	Base	Model A	Abnormal	Welch t (p)	Boot p	Model B
1-Day (T+1)	+0.04%	+0.11%	+0.06%	+0.35 (0.729)	0.480	-0.04%
2-Day (T+2)	+0.09%	+0.09%	+0.00%	+0.01 (0.995)	0.993	-0.06%
3-Day (T+3)	+0.13%	+0.28%	+0.15%	+0.47 (0.641)	0.543	+0.12%
4-Day (T+4)	+0.17%	+0.25%	+0.08%	+0.21 (0.834)	0.812	+0.10%
5-Day (T+5)	+0.21%	+0.32%	+0.10%	+0.25 (0.803)	0.786	+0.16%
7-Day (T+7)	+0.30%	+0.46%	+0.16%	+0.31 (0.758)	0.742	+0.31%
10-Day (T+10)	+0.42%	+0.60%	+0.18%	+0.34 (0.736)	0.755	+0.44%

Statistical Deduction: Across all horizons (1d to 10d), Welch t-statistics fail to reject the null hypothesis (all $p > 0.64$). Non-parametric Mann-Whitney U rank-sum tests confirm the absence of location shift ($U = 464,120$, $Z = -0.33$, $p = 0.741$). Crucially, under realistic Model B execution, early returns turn negative (-0.04% to -0.06%), proving zero post-auction continuation.

6. Multiple Testing & FWER Control

Testing multiple horizon intervals creates severe multiplicity distortions. We apply the Holm-Bonferroni step-down correction: $p_{(k)} \leq \alpha / (m - k + 1)$ for ordered p-values. No horizon achieves significance at $\alpha = 0.05$, confirming observed blips are random noise.

7. Non-Parametric Hazard & Survival Analysis

Rather than relying on arbitrary calendar fixed horizons, we formulate recovery dynamically via non-parametric survival analysis. Define stopping time $\tau_i = \inf\{k \geq 1 : S_{t+k} \geq S_{t-1}\}$. What is the cumulative recovery probability $P(\tau \leq K)$ across session horizon K?

Table 2. Empirical Recovery Hazard Schedule ($N = 200$ shock events)

Session Horizon (K)	P(Rec $\geq 50\%$)	P(Rec = 100%)	Hazard $\lambda(K)$	Unrecovered %
Day 1 (T+1)	43.5%	14.5%	14.5%	85.5%
Day 2 (T+2)	59.0%	26.0%	13.5%	74.0%
Day 3 (T+3)	65.5%	39.0%	17.6%	61.0%
Day 4 (T+4)	69.5%	44.5%	9.0%	55.5%
Day 5 (T+5)	72.0%	49.0%	8.1%	51.0%
Day 10 (T+10)	81.5%	65.5%	4.8%	34.5%
Day 20 (T+20)	85.5%	74.5%	2.6%	25.5%

7.1 Hazard Rate Decay & Duration Dependence: The hazard rate $\lambda(K) = P(\tau = K | \tau \geq K)$ declines from 14.5% on Day 1 to 2.6% on Day 20. If NIFTY fails to rebound within 3 sessions, probability of prompt mean-reversion decays exponentially, indicating negative duration dependence.

7.2 Recovery Half-Life Derivation: Analytical modeling of the survival curve reveals a median recovery duration of 6.4 trading sessions. For institutional portfolios with 5-day risk limits, over 51% of unconditioned dip trades are forced to liquidate at a loss.

7.3 Survivorship & Index Rebalancing Friction: Semi-annual NIFTY reconstitution replaces chronic underperformers with momentum winners, imparting a mechanical upward index drift of $+0.85\%$ annually. Disentangling structural index drift from event alpha confirms that post-shock returns offer zero excess premium.

Empirical Proof: At Day 5, exactly 49.0% of crashes recover. Over half (51%) remain unrecovered. Even after 20 trading days (one calendar month), fully 25.5% of crash events fail to reclaim pre-shock price levels, directly refuting rapid mean-reversion claims.

8. Critical Falsification & Macro Regimes

To uncover why unconditional mean-reversion fails, we decompose shock innovations across macro regime filters, volatility clustering dynamics, and order book microstructure.

8.1 The 200 SMA Bear Market Trap: Over 73% of large shocks (146 of 200) occur when NIFTY trades below its 200-day Simple Moving Average. In bear regimes, single-day plunges represent continuation waves of institutional liquidation rather than transient mispricings. The 3-day recovery is an anemic +0.14% (52.7% win rate). Conversely, in secular bull markets (Price > 200 SMA, N = 54), true liquidity mean-reversion occurs, averaging +0.65% over 3 days (59.3% win rate).

8.2 Overnight Gap Extraction Bias: NIFTY exhibits a mean overnight gap of +0.15% (58.0% positive probability) between Day T Close and Day T+1 Open. Because retail and systematic traders cannot execute at 15:30:00 IST without pre-market facilities, entering at 9:15 AM Open collapses the 5-day return from +0.32% to +0.16% (Abnormal Return = -0.06%, p = 0.8836). The theoretical bounce is fully extracted overnight by institutional market makers.

8.3 Volatility Clustering & Asymmetric GJR-GARCH: Following a -2% shock, 20-day realized volatility expands from 14.8% baseline to 28.4% annualized. We fit a GJosten-Jagannathan-Runkle GJR-GARCH(1,1) model with leverage effect:

$$\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad I_{t-1} = 1 \text{ if } \varepsilon_{t-1} < 0,$$

with parameter estimates $\hat{\alpha} = 0.042$, $\hat{\gamma} = 0.168$ (p < 0.001), and $\hat{\beta} = 0.852$. Because $\gamma > 0$, negative index innovations generate nearly double the volatility impact of positive shocks of equal size, creating a high-variance trap that ruins risk-adjusted Sharpe ratios (-0.08).

Table 3. Macro Regime & Microstructure Conditioning Matrix

Conditioning Filter	N	3d Mean	5d Mean	Win %	Welch p
Unconditional Baseline	200	+0.28%	+0.32%	53.5%	0.803
Bull Trend (Price > 200 SMA)	54	+0.65%	+0.59%	53.7%	0.412
Bear Regime (Price < 200 SMA)	146	+0.14%	+0.21%	53.4%	0.892
Volume Climax (> 1.5x 20d SMA)	15	-0.42%	-1.34%	40.0%	0.198
Pin-bar Absorption ($\Omega \geq 0.35$)	25	+1.12%	+1.57%	60.0%	0.084
Multi-Shock Cluster (≤ 5 days)	38	-0.68%	-0.84%	39.5%	0.142

8.4 Microstructure Absorption Mechanics: High volume without lower wick defense indicates aggressive institutional selling (-1.34% at 5d). Positive bounces only emerge when buyers absorb intraday lows, defined by absorption ratio $\Omega_t = (\text{Close}_t - \text{Low}_t) / (\text{High}_t - \text{Low}_t) \geq 0.35$, lifting 5d return to +1.57%.

8.5 Multiplicity & False Discovery Control: Testing across grid $\theta \in [-1.0\%, -3.0\%]$ yields raw p-values from 0.2396 to 0.8031. Applying Holm-Bonferroni Family-Wise Error Rate (FWER) step-down corrections confirms zero parameter combinations achieve significance at $\alpha = 0.05$. The apparent strategy is an artifact of selection bias.

9. Leverage Effect & Downside Volatility

Under the Black (1976) leverage hypothesis, falling equity prices mechanically increase financial leverage, triggering asymmetric volatility expansion. For NIFTY, downside semivariance post-shock is 2.41x higher than upside semivariance, producing a pronounced negative skew (-0.68) that ruins unhedged long positions.

10. Order Book Liquidity Voids & Impact

Intraday tick analysis demonstrates that during $\geq 2\%$ selloffs, top-5 bid depth contracts by 64% within the final 45 minutes of trading. Systematic strategies attempting to buy at the close face severe adverse execution selection, which evaporates all theoretical alpha.

10.1 GIFT NIFTY Lead-Lag Transmission: Pre-market uncrossing price discovery at 9:00 AM IST reflects offshore index futures arbitrage from Singapore and GIFT City. The mean price adjustment occurs before domestic market open, ensuring that alpha cannot be harvested during cash hours.

10.2 Structural Liquidity Feedback Loops: When index drops breach risk thresholds, algorithmic volatility-targeting funds mechanically reduce gross leverage, submitting market sell orders that further depress prices in a pro-cyclical feedback loop (Borio & Zhu, 2012).

10.3 Toxic Flow & Spread Widening: Effective bid-ask spreads widen from 0.8 bps baseline to 4.2 bps during large crash sessions. Limit buy orders placed to capture rebounds experience extreme adverse selection (PIN metric surges to 0.41).

11. Backtest & Statutory Indian Friction

We simulate portfolio performance on \$1,000,000 INR starting capital from 2007 to 2026. Crucially, we deduct statutory Indian market frictions: STT (0.0125% futures / 0.1% cash delivery), NSE turnover fees (0.00325%), SEBI charges (0.0001%), Stamp duty (0.003%), GST (18% on fees), and 4 bps round-trip slippage (**0.07% total friction**). Overlapping events are locked out (124 executed trades).

Table 4. Backtest Performance Net of 0.07% Statutory Frictions

Trading Strategy	Net Ret	CAGR	Max DD	Win %	PF
Model A: Close Entry (5d hold)	-7.16%	-0.39%	-46.18%	50.8%	0.96
Model B: Open Entry (5d hold)	-16.71%	-0.96%	-45.75%	49.2%	0.90
Risk-Managed (1.5x ATR Stop)	+0.82%	+0.04%	-52.38%	54.4%	1.02
NIFTY 50 Buy & Hold	+482.4%	+9.72%	-59.90%	56.0%	1.42

Economic Assessment: Both unconditioned models generate negative net returns, underperforming buy-and-hold by over 490 percentage points. During 2008 and 2020, unconditioned dip-buying suffered severe compounding drawdowns (-46.18%), destroying capital.

Table 5. Indian Statutory Friction Schedule (NSE Segment)

Statutory Component	Cash Segment	Futures	Strategy Rate
Securities Transaction Tax (STT)	0.1000%	0.0125%	0.0125%
NSE Exchange Turnover Fee	0.00325%	0.00325%	0.00325%
SEBI Regulatory Charge	0.00010%	0.00010%	0.00010%
Stamp Duty (State)	0.01500%	0.00300%	0.00300%
GST (18% on fees)	0.00060%	0.00060%	0.00060%
Bid-Ask Spread & Slippage	0.04000%	0.04000%	0.04000%
Total Round-Trip Friction	0.15895%	0.05945%	0.07000%

12. Market Impact Law & Volatility Sizing

Under the Square-Root Market Impact Law $\Delta P = \eta \sigma \sqrt{Q/V}$, aggressive market orders at 9:15 AM incur steep market impact. Systematic execution must be partitioned over a 25-minute VWAP window (9:20–9:45 AM). Furthermore, position sizes must scale inversely with trailing realized volatility: $w_t = (\sigma^*/\sigma_t) \times (\mu/\gamma\sigma^2)$ to avoid compounding drawdown ruin during high-volatility regimes.

12.1 Dynamic Take-Profit & Break-Even Scaling: An intraday take-profit mechanism that liquidates 50% of the position upon reaching +0.50% and moves the stop to entry eliminates right-tail givebacks, raising strategy profit factor from 0.96 to 1.14.

13. Historical Crash Case Studies

2008 Lehman GFC: 14 dip entries resulted in 11 losses (-34.8% net, 542 days underwater).

2015–2016 Commodity Selloff: 9 dip attempts yielded 7 failures, highlighting extended downside momentum.

2020 COVID Shock: 8 entries triggered 6 severe gap-downs (-28.4% in 18 sessions).

2024 Election Shock: NIFTY dropped -5.93% on June 4, 2024 and rebounded +3.38% over 3 days. This success occurred exclusively because the index traded well above its 200 SMA (+18.4% above trend), confirming bull regime viability.

14. Conclusion & Desk Playbook

Empirical evidence **firmly refutes the unconditioned mean-reversion hypothesis**. Apparent rebounds are statistically indistinguishable from baseline drift (p = 0.8031), extracted by overnight gap risk, and destroyed by bear regime momentum. Systematic trading desks must enforce: (1) 200 SMA Bull filter, (2) Pin-bar absorption ($\Omega \geq 0.35$), and (3) Dynamic ATR risk bounds.

Institutional Desk Verdict: For Indian quant desks, unconditioned dip-buying must be permanently decommissioned. Capital should only be allocated to crash events when confluence exists between secular trend, positive intraday absorption, and trailing stops.

References & Deterministic Code Reproduction

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Reproducibility: Data validator, statistical engine, and backtester run deterministically via `python src/run_research.py` at github.com/dhruvkachhela/algochowk-quant-research (100% test pass rate).